Application architecture

Layered structure, module responsibilities and data flow

Sheet 2

Format A1

Layered architecture

Presentation layer

- index.html — single-page client
- app.js — map & controller
- api.js — REST + WebSocket
- charts.js — metrics plots

HTTP / WS

Service layer (FastAPI)

- REST endpoints — /api/route, /api/classify
- WebSocket /ws/plan — live progress streaming
- Tile service — on-board chart

calls

Domain layer

- ZoneClassifierService — inference façade
- RoutePlanner — orchestration
- CostMap — lattice + weights
- A* / Dijkstra — graph search

reads

Data layer

- ais_loader — GeoJSON/CSV/JSON
- features — 28 engineered inputs
- ReferenceIndex — BallTree, label-free
- repository — SQLite persistence

writes

Export layer

- GeoJSON writer
- CSV writer
- PDF report (ReportLab)

Request sequence of one planning job

1. client sends request
2. build lattice
3. network classifies cells
4. cost map + safety margin
5. A* / Dijkstra search
6. compare with great circle
7. persist + stream result

Progress frames (stages 2–5) are pushed to the browser over the WebSocket, so the operator watches the route being built rather than waiting for a single response.

Database schema (SQLite)

session
- id PK
- name, source
- n_points
- created_utc

1 : N

point
- id PK
- session_id FK
- latitude, longitude
- depth_m, elevation_m
- class_code FK
- confidence

FK → **zone**
- code PK
- title, description
- traversal_cost
- navigable

1 : N

route
- id PK
- route_uid UNIQUE
- algorithm
- start/end lat, lon
- total_cost, distance_km
- great_circle_km, detour_ratio
- nodes_expanded, runtime_s

1 : N

waypoint
- id PK
- route_id FK
- seq
- latitude, longitude
- class_code FK
- cumulative_km

Foreign keys are enforced (PRAGMA foreign_keys = ON); the view v_route_summary aggregates routes with their waypoint counts.

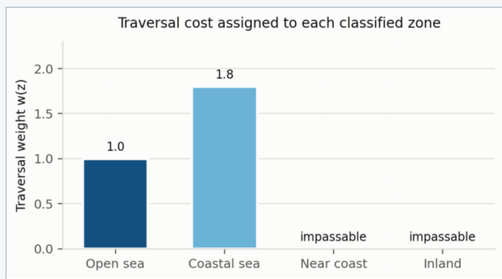
Classification model

Residual multilayer perceptron, 28 inputs → 4 classes

- Hidden layers 256 – 128 – 64, GELU, batch norm, dropout 0.2
- 98,364 trainable parameters
- AdamW, lr 0.0015, batch 512
- Trained 97 epochs in 4.3 min on CPU

Feature groups

- Geodetic encoding (4) · point bathymetry (5)
- Neighbourhood context (9) · OSM objects (10)



Application for constructing a maritime vessel's course based on neural-network classification of geographic zones

Diploma project · Student: Li Haoran

Brest State Technical University · Faculty EIS, Department IIT · 2026/27